

Clinical Geriatric Functional Assessment using Wearable Sensing and Machine Learning

Rishank Singh¹, Sankalp Vaish¹, Xiaohui Liang¹, Jennifer G. Poole², and John A. Batsis², Danae C. Gross²

¹Department of Computer Science, University of Massachusetts Boston, MA, USA

²School of Medicine, University of North Carolina at Chapel hill, NC, USA

Email: {Rishank.Singh001, Sankalp.Vaish001, Xiaohui.Liang}@umb.edu,

Jennifer_Poole@med.unc.edu, {John.Batsis, dcgross}@unc.edu

Abstract—Physical function is critically important for older adults as it directly impacts their independence, quality of life, and overall health. To assess the physical functions of older adults, primary health provider are optimally placed, but they do not complete the assessments due to time and space constraints. In this paper, we introduce a geriatric functional assessment (GFA) system using a wearable device and machine learning methods to assess older adults' physical function during their clinical visits and enable clinicians to monitor physical function results. Specifically, the GFA system includes a badge-sized wearable prototype with motion and visual sensors that continuously collect sensor data during ~15-minute clinical visits. Then, the GFA system integrates three different physical function inference methods using a support vector classifier, random forest, and deep neural network to infer actionable clinician-familiar measures, including gait speed, sit-to-stand, and balance metrics. We tested the GFA among 21 older adults who visited the same clinic. The evaluation of the inference methods shows that the random forest classifier performs better than the Support vector classifier and Dense neural network; the accuracy score is increased from 68.80 to 74.10 for gait speed, from 60.70 to 66.90 for sit-to-stand and from 69.80 to 76.2 for balance. Moreover, we showed that the GFA's usability score is promising, 79 out of 100, and we identified critical design issues that should be addressed in future designs.

Index Terms—Physical function, older adults, wearable sensors, motion and visual, machine learning

I. INTRODUCTION

Maintaining physical function is crucial for older adults' independence and health, impacting their daily activities and linking to frailty, disability, and mortality. Primary care providers can screen for early signs of decline, but challenges like limited time, insufficiently trained staff, and reimbursement issues hinder regular assessments. Similarly, self-reported surveys can be utilized but vary in completion and accuracy compared to validated measures.

Wearable sensors can obtain real-time activity data in a continuous, unobtrusive, and validated manner. Researchers have explored wearable solutions for the healthcare domain in a free-living setting and a controlled setting. In the free-living (home) setting, wearables are used to recognize activity types, detect falls, and track long-term step counts [1], [6], [10], [13], [18], free-living sensor data is ineffective in inferring accurate physical function, such as gait speed, sit-to-stand,

and balance, that are often reviewed by clinicians. The challenges are associated with complex sensing environments (e.g., privacy concerns of continuous visual recording at home), different walking patterns, different footwear, and unknown health conditions related to diet or medication changes, which largely impact the inference accuracy. In controlled settings – research laboratories and rehabilitation centers, treadmills [9], [17], floor sensors [11], [23] or virtual reality headsets [7], [16] can assess physical function [8]. Specifically, users perform tasks (e.g., Timed Up and Go [21], Clinical Test of Sensory Integration for Balance [5]) that are supervised by trained staff while sensors on these devices record the motion data, which is then automatically analyzed for physical function assessments. The process held in controlled settings minimizes the human effort and time required to assess measures but still requires personnel, space, and time to conduct tasks. In this paper, we aim to propose a new physical function assessment system that is designed to be used in a clinical environment, which is neither a free-living nor a controlled setting. The system is expected to achieve high practicality, usability, and accuracy by taking advantage of older adults' spontaneous movements in a clinical environment and structured clinical activities [2], [14].

We propose a clinical geriatric functional assessment (GFA) system that consists of a badge-sized wearable prototype, a data collection protocol, and physical function inference methods. The goal of the GFA system is to collect and analyze sensor data to infer physical function values, including gait speed, sit-to-stand repetitions, and standing balance metrics. The GFA system works assuming a high correlation between wearable sensor data and physical function. We envision three distinct advantages of the GFA: (i) periodic collection in clinics with similar setups, enabling comparable and reliable data collection over time; (ii) generated from spontaneous movements as older adults move about the clinic, thus incurring minimum effort and enabling objective assessment; and (iii) time-aligned with clinical data (e.g., vital signs) of each visit. Technically, the GFA is envisioned to achieve high inference accuracy by jointly analyzing time-series motion and visual data. Visual data analysis relies on distinctive reference objects to provide contextual data, including location and proximal physical objects (e.g., chairs, doors, tables). Motion data are directly related to physical movements but lack

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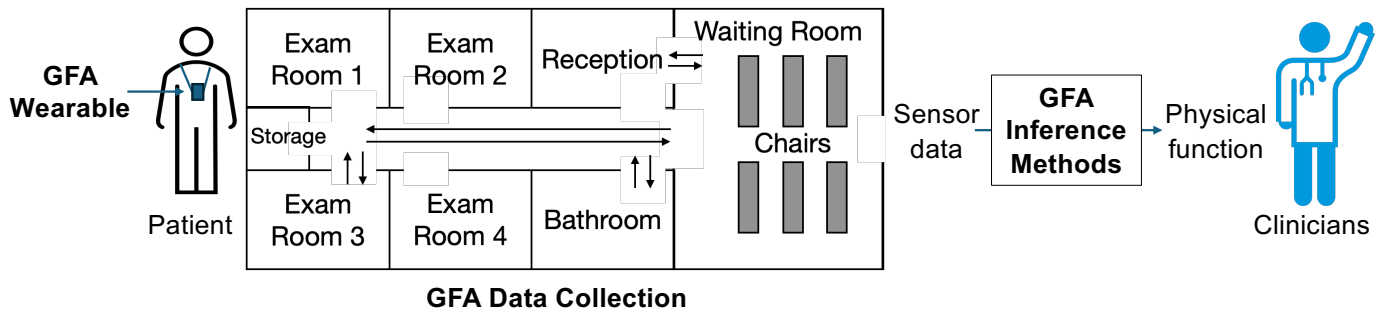


Fig. 1: Three components of GFA: GFA wearable prototype, GFA data collection protocol, and GFA inference methods

context (e.g., walking while turning, sitting on chairs that have different heights), which is not reliable for accurate inference of physical function. Thus, the GFA analyzes motion data for its high relevance to physical movements and visual data for the environmental context to produce accurate, detailed, and interpretable results. Our contributions are three-fold:

First, we design a GFA wearable prototype and a GFA data collection protocol to enable motion and visual data collection during an older adult's clinical visit. The GFA prototype reliably collected data on 21 older adults in a real-world clinic.

Second, we propose three GFA inference methods to infer physical function values from the sensor data. We propose the support vector classifier (SVC), random forest (RF), and deep neural network (DNN) methods to analyze motion data.

Third, we assessed the accuracy of each model by varying the window size of the data. We confirmed through our experiment that the window size of 15 outperforms the other window size of 10 20 and 30 by increasing the accuracy from 74.10 to 69.90 for gait speed, from 53.90 to 66.90 for sit-to-stand and from 63.40 to 76.20 for balance.

II. GFA WEARABLE PROTOTYPE

A. Hardware Setup

We have assembled sensors and 3D-printing cases as shown in Fig. 2(a). The GFA wearable prototype has three major components: visual, motion, and battery components.

Visual component. We have used two high-featured sensor boards from Portenta, i.e., Portenta H7, along with a vision shield. The vision shield comes with a 320x320 pixels camera sensor, which we used to collect a series of images. The Portenta H7 allows for programming with high-level languages and AI and has highly customizable hardware. Both Portenta H7 and Vision Shield are highly compatible with each other, which makes our wearable prototype more efficient and fast.

Motion component. We have chosen Arduino MKR zero along with Arduino IMU shield. MKR Zero comes with an SD card holder and dedicated SPI interfaces. IMU shield read three-dimensional acceleration, yaw rate, and magnetical field.

Battery component. The GFA wearable prototype needs to be powered on for each older adult's visit. We consulted with the clinicians and staff of our team and found the clinical visit duration is no longer than 30 minutes. Thus, we tested the GFA sensing for 30 minutes, and the battery consumption

was around 5 mAH. We chose a portable battery with 10,000 mAH due to its cheap price, slim design, and large capability.

Integration and testing. Throughout our study, we made several modifications to the integration process to optimize its performance and functionality. These modifications include changes in the collection frequency of motion and visual data, as well as the coordination to process and transfer data. Moreover, we developed a comprehensive document outlining the hardware specifications, wiring diagrams, and operational procedures to ensure reproducibility and facilitate future research endeavors.

III. GFA DATA COLLECTION

The GFA data collection was conducted in a clinical environment where older adults wore the GFA wearable prototype and moved spontaneously during their clinic visits. The data collection protocol has **five** steps.

Setup device: A clinical staff arrived 10 minutes prior at the check-in place to meet with the participant. The RA went over the details of the study and the purpose of the study, and gave the participants a consent form with every detail of the study. After getting the consent form, the staff helped put the device on the participant. A light strap was connected to the device, and the participant wore it around the neck so that the GFA device would not fall. The RA used a clip to fix the device on the participant's T-shirt so that it would not flip. A battery was connected to the device and put in the pocket.

Record while moving: While wearing the GFA device, the participant completed a clinical workflow as shown in Fig. 2(b). These activities include walking around the hallways, toward the waiting room, and later to the exam room with staff guidance. This process takes about 15 minutes.

Remove device: After the participant enters the exam room, the staff removes the device from the participant, stops data collection, transfers the data from the SD card to the computer, empties the SD card, and sanitizes the devices. The staff also cleaned and sanitized any surfaces and objects that may have been touched during the visit.

Measures: The staff conducted conventional physical function measures on gait speed, sit-to-stand, and postural control. *Gait speed* - walk 4 meters, and the average time to complete 3 rounds is recorded. *Sit to stand* - repeatedly sit and stand for 30 seconds, and the number of repetitions is recorded. *Postural*

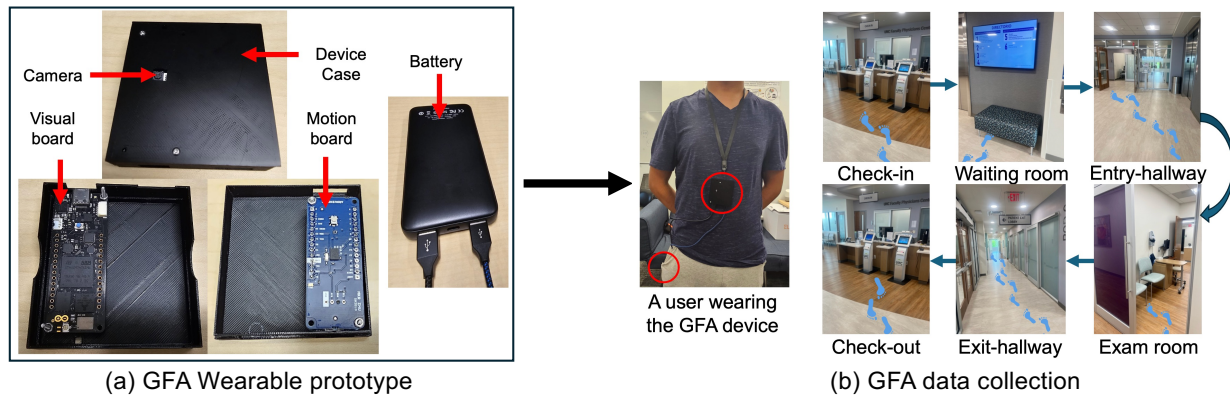


Fig. 2: GFA wearable prototype and GFA data collection

control - stand on a pressure plate in two conditions of eye-opened and eye-closed, and the moving areas and distances of the center of pressure are recorded.

Exit surveys and interviews: Participants completed the usability surveys in REDCap on the computers. They further answered questions in an exit interview where their audio was recorded. Finally, they received a \$20 gift card as an appreciation token for their participation.

IV. DATASET

We conducted the GFA data collection on 21 older adults and obtained the motion, visual, and physical function datasets.

Motion dataset. The motion sensor on the GFA wearable prototype records acceleration using a triaxial accelerometer along with magnetic field, gyroscope, and Euler angles. On average, the device recorded 4,113 samples at a consistent rate of 4.4 samples per second. As shown in Fig. 3, Participant P9 recorded 6,102 samples (largest) over 1,398 seconds, significantly longer than the average duration of 21 participants (893 seconds). In contrast, P7 recorded 2,886 samples (smallest) over 661 seconds (shortest). Overall, the device performed exceptionally well in gathering motion data as the sample rate was almost constant at 4.4 samples per second for all the participants, which demonstrates the GFA's reliability in motion data collection.

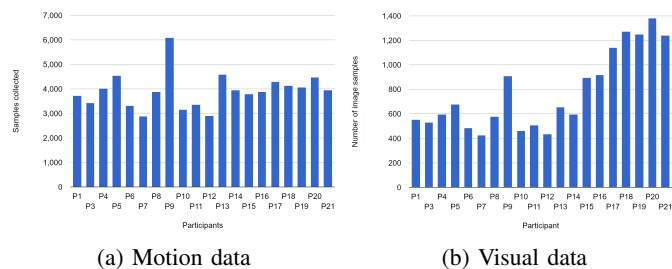


Fig. 3: Motion and visual datasets

Visual dataset. Compared to motion data collection, we encountered more challenges in visual data collection. On average, GFA collects 776 images per participant with a

sampling rate of 0.9 images per second. Initially, we found a sampling rate of 0.64 images per second for early participants, and later, we upgraded the SD cards with a faster transfer rate and significantly increased the frequency to 1.34 images per second. As shown in Fig. 3b, we have P1-P14 records 0.6 images per second, P15, P16 records 1.0 images per second, and P17-P21 records 1.34 images per second. Our proposed inference methods are designed to accommodate visual data at different sampling rates.

Physical function dataset. Our staff supervised participants to complete three physical function measures after the GFA data collection. As shown in TABLE I, the average gait speed was 0.88 meters per second (m/s). Among them, P2 had the slowest gait speed at 0.589 m/s, whereas P12 achieved the fastest gait speed at 1.25 m/s. For the sit-to-stand repetitions, P13 and P21 excelled, completing more than 20 stands in 30 seconds. The staff also reported that two participants experienced imbalance, and five participants relied on their arms to sit or stand up from a chair. For the balance, when participants had their eyes open, the average ellipse area covered by the center of pressure was 24.48 mm², and the average distance traveled was 130.56 mm. With eyes closed, the average ellipse area decreased to 22.1 mm², while the average distance traveled increased to 204.92 mm [4], [15]. In summary, we observed that the participants had shown varying levels of physical function abilities.

Physical function	Mean	Highest	Lowest
Gait Speed (m/s)	0.89	1.32	0.51
Sit-to-stand repetitions	11.38	24	5
PC: EA, Open (mm ²)	24.48	70	1
PC: EA, Close (mm ²)	22.10	119	1
PC: TD, Open (mm)	130.56	493	38
PC: TD, Close (mm)	204.92	1556	6

TABLE I: Physical function dataset. PC: Postural Control. EA: Ellipse Area. TD: Travel Distance.

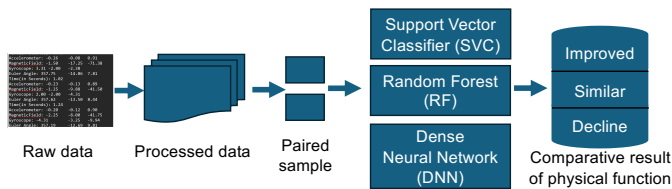


Fig. 4: GFA inference methods

V. GFA INFERENCE METHODS

We employed a combination of data analysis techniques and machine learning algorithms to process and analyze the motion and visual data collected from the device to derive a comparative result of physical functions. These comparative results of physical function can be derived from a pair of data samples and classified into three classes - improved, similar, and decline, if the second sample is in an improved, similar, or decline condition. Improved means faster gait speed, more sit-to-stand repetitions, and smaller travel distances. We leveraged machine learning algorithms, such as SVC, RF, and DNN, to show the comparative results based on the sensor readings. By training these models on labeled data, we were able to achieve high accuracy in inferring function changes.

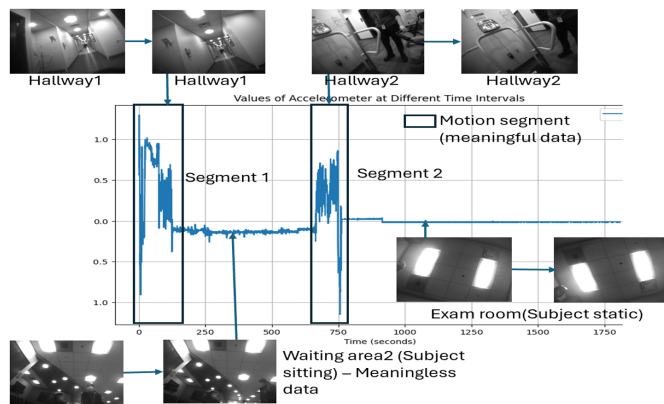


Fig. 5: Data Preprocessing

Data preprocessing - We meticulously filtered meaningful from meaningless data to ensure high-quality input for analysis. Initially, we identified and extracted two segments from the entire motion data, as shown in Fig. 5 where the participant was actively in motion, eliminating any stationary intervals. We found each participant generated two segments following the same clinic flows as shown in Fig. 5. We subsequently processed the motion data by selecting segments corresponding to 10 consecutive images, creating a series of sample data points. Each data window comprised around 40 motion data points. Each sample pair was formed from these windows, and duplicate pairs were excluded to maintain dataset integrity. This approach allowed us to count the number of unique sample pairs accurately. These pairs were subsequently used to train and test our model, ensuring a robust and reliable

performance evaluation. We then repeated the experiment for window sizes of 15, 20, and 30.

To further enhance the performance of our models, we implemented the sliding window approach. This technique involves segmenting the continuous motion data into overlapping windows, with each window capturing a portion of the data for analysis. Specifically, we set the step size of 5 and 10 to create these overlapping segments.

Data preprocessing. We utilized image data to confirm the activity context. We trained a classifier to categorize input images into one of five specific locations: waiting_area1, hallway1, waiting_area2, hallway2, and exam room, as shown in Fig. 5. As each data point consists of consecutive motion data samples, we confirm one activity context from the associated images. This activity context was then combined with motion data to train our models. We specifically leveraged motion data collected in “moving from reception to waiting area” and “moving from waiting area to exam room,” as these activities capture participants in motion, providing valuable data for inferring physical functions. In contrast, data from other activities “sitting in the waiting area” were further confirmed to be meaningless via the images and eliminated from the analysis.

Feature extraction. We utilized accelerometer, gyroscope, Euler angle, and magnetic field data to assess physical functions such as gait speed, balance, and sit-to-stand transitions. For feature extraction, we calculated the mean of the XYZ axes for each metric of all the motion data obtained in a window. The rationale for using the mean feature is grounded in its ability to provide a stable and representative value of the overall movement, reducing the impact of transient noise and fluctuations [24].

Models. We implemented the following models:

A **Support Vector Classifier (SVC)** was ideal for our research because it is highly effective in high-dimensional spaces, robust to overfitting especially in smaller datasets by maximizing the margin, and versatile with its ability to handle both linear and non-linear classification problems through the use of kernel functions [3] [12].

A **Random Forest (RF) model** is robust against overfitting and it can handle large datasets with high dimensionality, and its capability to provide feature importance ranking, aiding in insightful interpretation of results and understanding of underlying relationships in the data [20] [22] [19].

A **Dense Neural Network (DNN) model** demonstrates superior adaptability to complex datasets by leveraging deep architectures. Additionally, their inherent capacity for automated feature extraction eliminates the reliance on handcrafted features, streamlining the modeling process compared to traditional methods. Moreover, the ability of neural networks to learn hierarchical representations enables them to capture intricate relationships within data [25].

VI. EVALUATION

In this section, we evaluate the performance of the inference methods, including SVC, RF, and DNN. We will compare their

effectiveness using motion data with features and additional information extracted from image data. The following analysis also highlights the usability and acceptance of GFA device, along with the comparative results of different window sizes.

A. Usability and acceptability

The usability of the GFA device has proven to be 79.29 which is higher than expected, as evidenced by user feedback during interviews. For instance, participant 18 highlighted how easy it is to operate the device without extensive training. Similarly, Participants 8 and 1 appreciated the design of the device as it was very comfortable to use and it just needed a strap to wear around the neck. However, the acceptability of the device was 68.30 lower than anticipated, as several users pointed out notable limitations. Most of the users were skeptical about the full potential of the device. Participant 19 in particular highlighted that the device would be much more acceptable if the device had a handout along with it or a manual describing its functionality. Participant 14 mentioned that despite the device's functionality, its appearance is too much like a prototype, reducing its practicality in fieldwork. Additionally, Participants 1 and 11 expressed concerns about the high cost and said they would buy it only if it was low cost or covered under medical insurance. Overall, while the device excels in usability, making it easy and efficient to use, its acceptability is hindered by issues related to cost and compatibility. These insights suggest areas for improvement to enhance the device's overall adoption.

B. Inference accuracy across models

Window	Model	Sit-to-stand	Gait-speed	Balance
10	SVC	57.9	68.4	65.1
	RF	63.5	72.7	69.4
	DNN	62.1	69.8	64.3
15	SVC	60.7	68.8	69.8
	RF	66.9	74.1	76.2
	DNN	61.8	73.3	71.3
20	SVC	62.1	66.9	68.7
	RF	57.9	73.5	64.3
	DNN	57.9	71.7	68.5
30	SVC	57.5	69.2	54.9
	RF	53.9	69.9	57.5
	DNN	56.8	72.5	57.5

TABLE II: Accuracy for models and non-overlapping windows

We evaluated the performance of three machine learning models—SVC, RF and DNN—using accuracy as the evaluation metric as shown in Table II. Our goal was to identify which model provided the most accurate inference based on the features extracted from motion and image.

For a window size of 10, the RF model outperformed all others in Gait Speed (72.7%), Balance (69.4%), and Sit-to-Stand (63.5%). With a window size of 15, RF again maintained superior performance in Gait Speed (74.1%), Balance (76.2%), and Sit-to-Stand (66.9%). For window size 20, RF was most accurate in Gait Speed (73.5%) SVC model showed the best accuracy in Sit-to-Stand (62.1%) and Balance (68.7%). Lastly,

with a window size of 30, DNN achieved the highest accuracy across Gait Speed (72.5%), SVC for Sit-to-Stand (57.5%), and Random Forest for Balance (63.4%). This comparative analysis highlights the varying performance of different models and shows accuracy decreases as window sizes increase, with RF generally showing higher accuracy across all window sizes. These results highlight the efficacy of the Random Forest model, particularly with a window size of 15, across different physical function assessments.

RF can automatically determine the importance of different features. This is particularly beneficial for motion data, where the relevance of specific features may vary across different tasks (e.g., gait speed vs. balance). Motion data includes time-series data that exhibits complex, non-linear patterns. Random Forests can handle these non-linear relationships effectively by using multiple decision trees, each focusing on different aspects of the data. DNN was prone to overfitting as the dataset contained pairs of the same sample, not enough to support the model's complexity. RF, through the averaging of multiple trees, is less likely to overfit the training data. Motion data can be noisy due to various factors, such as sensor inaccuracies or environmental variations. RF is inherently robust to noise since it aggregates the results of multiple decision trees, each of which might be trained on a slightly different subset of the data. Neural Networks are more sensitive to such noise.

C. Inference accuracy by window size and sliding window

We tested four different window sizes: 10, 15, 20 and 30 as shown in Table. II. Window size can be defined as the number of images and corresponding motion data points grouped for analysis. For example, given 4000 corresponding motion data points, a window size of 10 means that the data is segmented into 100 windows, each containing 10 images and 40 corresponding motion data points. Our goal was to evaluate how the window size of motion data would impact the model's accuracy and capabilities.

The impact of window size on the performance of our machine-learning model is multifaceted and context-dependent. Increasing the window size provides more contextual information, which can help the model capture comprehensive patterns and dependencies. However, too big window size increases the risk of underfitting as it decreases the number of samples. Conversely, decreasing the window size allows the capture of fine-grained temporal patterns, but it may lead to higher computational costs and increased sensitivity to noise. Given these considerations, the optimal window size must be determined experimentally, taking into account the specific characteristics of the data and the problem at hand. In our case, the most optimal window size was 15.

To further enhance our results, we implemented the concept of sliding windows, using a step size of 5 and 10. This approach significantly increased the amount of training data. Our experiments demonstrated that this method improved the accuracy of every model. Furthermore, for all models (SVC, RF, and DNN), a smaller sliding size (5) generally yields higher accuracy compared to a larger sliding size (10).

Notably, the accuracy for all physical functions, even for Sit-to-Stand, exceeded 87% for window size 30 and step size 5, as shown in Table III, whereas it did not even reach 70% without using the sliding window technique.

Model	Window	Sliding	Sit-to-stand	Gait speed	Balance
SVC	10	5	64.9	73.3	69.5
	15	5	63.4	75.2	75.1
	20	5	67	70.5	72.9
	30	5	67.6	72.3	73.9
	10	10	57.9	68.4	65.1
	15	10	53.1	65.6	67
	20	10	62.9	70.2	66.1
	30	10	68.5	71.7	72.2
RF	10	5	75.8	80.7	77.8
	15	5	75.2	82.3	80.7
	20	5	82.2	86.1	83.7
	30	5	87.5	87.6	88.7
	10	10	63.5	72.2	69.4
	15	10	62.2	71.2	69.1
	20	10	67.8	79.4	72.3
	30	10	72.4	75	78.6
DNN	10	5	68.7	75.4	72.1
	15	5	65.9	75.6	77
	20	5	74.7	76.1	79.2
	30	5	76.6	79.5	68.5
	10	10	62.1	69.8	64.5
	15	10	56.6	71.3	68.8
	20	10	69.2	78	70
	30	10	69.7	75	75

TABLE III: Accuracy for models, window, and sliding

The sliding window approach significantly enhances model performance by creating a more diverse and comprehensive training dataset. By overlapping windows, we capture more temporal variations and subtle changes in motion data, which are crucial for accurately estimating physical functions.

The table III illustrates the phenomenon where smaller sliding sizes lead to higher accuracy metrics, potentially due to the redundancy in overlapping samples. Using a shorter step size of 5, the similarity within samples increases as we are using the mean of the motion data. This causes more bias between the training and test sets, which may artificially inflate accuracy, as shown in the table. For instance, with a window sizes of 30 and 20 and a step size of 5 (highlighted in the table), the mean values will be very similar across samples, resulting in the highest accuracy. Same phenomenon is also shown by DNN model with window size of 20 and 30 and step size 5. This result indicates that while shorter step sizes can improve accuracy, they may do so by introducing bias rather than by truly enhancing model performance.

While the sliding window approach has demonstrated significant improvements in model accuracy, it is essential to address the potential for data leakage, where overlapping data segments could inadvertently be used in both training and testing phases. This overlap can artificially inflate performance metrics, leading to overestimated model capabilities. In our future work, we plan to increase the number of participants to create a larger dataset, allowing us to divide the participants' data into distinct training and testing sets. This will help eliminate any bias introduced by the sliding window technique.

VII. DISCUSSION

We faced several challenges of the proposed design, method and context, and we received valuable insights for understanding the results and guiding future research.

Window techniques. We adopted window techniques by creating many samples. These samples are made of 10-30 images, and 40–120 motion samples. The number of samples is increased by shortening the window size, which would help us train the model. However, in reality, we may use the data collected from a much longer time period (e.g., a whole session of about 15 minutes) to infer the physical function. Removing the noise from the sensor data and extracting time-series features from multi-modal sensor data of a long duration are critical challenges. Research efforts are needed to address these challenges when more data is collected.

Design issues. The device was designed to in a way a that the participant do not feel minimal discomfort while wearing or taking it off. So we decided to hang it with a neck strap and do not add anything other than that. However there was some unexpected flipping of the device during the movement and this caused the camera to turn the other side and loss of data. In future research comfort as well as these unexpected accidents that may happen should also be considered.

Person with walker. During our analysis, we figured out from the visual data that two older adults used walker while walking from one place to another. This can be an issue as the physical function test that was conducted separately without the use of walker. This inconsistency may cause the low accuracy of the model, and emphasize the importance of the visual data collection.

Visual data frequency. The sample rate of images in our visual data was improved throughout the experiment and still has the potential to be improved further. We currently have the visual frequency at 1.34. At such frequency, the visual data can be used to confirm pre-defined activity context as additional feature to the model but not used directly for physical function inference in the inference model. We envision in future testing we will obtain more frequent visual data and improve the inference accuracy by additionally calculating the pattern changes and automatic features of the time-series visual data.

VIII. CONCLUSION

We introduced and evaluated a clinical geriatric functional assessment (GFA) system using a wearable device and machine learning methods to assess older adults' physical function during their clinical visits. The GFA reliably collects data from older adults' spontaneous movement in the clinic and receives a promising usability score 79/100 from older adults. The analysis of the GFA inference methods confirms that random forest outperformed the SVC and deep neural networks methods by increasing the accuracy from 68.80 to 74.10 for gait speed, from 60.70 to 66.90 for sit-to-stand and from 69.80 to 76.2 for balance. We also confirmed based on our experiment that a window size of 15 is most optimal for our research. Through this study, we identify critical issues in device design, data collection, and inference methods. We plan

to conduct further studies to ensure both high-quality sensor data and maximize usability, which would enhance inference accuracy of the physical functions of older adults. Overall, this research underscores the importance of advanced data science research by developing a wearable prototype and applying deep neural network models that can provide clinicians with accurate data on changes in physical function and its potential to impact the physical functionalities of older adults.

IX. ACKNOWLEDGEMENT

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